

Code:

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//Program to find the maximum and minimum distances between any two points
//taken from a randomly given set of n distinct points

#include <iostream>
#include <cmath>
#include<iomanip>
using namespace std;

//#define MAX_ITEMS 1000

//Function to calculate distance between any two points
double disfun(double xl, double yl, double xr, double yr)
{
    double xd = xl - xr;
    double yd = yl - yr;
    return sqrt(xd * xd + yd * yd);
}

int main()
{
    cout<<"\nPlease make sure that all the points you enter are distinct\n";
    int i, j, n;
    double xleft, yleft, xright, yright;
    cout<<"\nEnter the number of points\n";
    cin>>n;

    if(n<2)
    {
        cout<<"\nThere should be at least two points\n";
        exit(0);
    }
    int N=n*(n-1)/2;
    double dist[N];
    double a[n][2];
    cout<<"\nEnter the points row-wise\n";
    for(i=0;i<n;i++)
        for(j=0;j<2;j++)
            cin>>a[i][j];

    //main algorithm
    int m=0;
    for(i=0;i<n-1;i++)
    {
        xleft=a[i][0];
        yleft=a[i][1];
        for(j=i+1;j<n;j++)
        {
            xright=a[j][0];
            yright=a[j][1];
            cout<<"("<<xleft<<","<<yleft<<")"<<endl;
            cout<<"("<<xright<<","<<yright<<")"<<endl;
            dist[m]=disfun(xleft, yleft, xright, yright);
            cout<<"\nThe distances are: dist["<<m<<"]"<<setw(5)<<dist[m]<<endl;
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        m++;
    }
}
//sorting the distances vector in ASCENDING ORDER
for(i=0;i<m;i++)
{
    for(j=i+1;j<m;j++)
    {
        if(dist[i]>dist[j])
        {
            double temp =dist[i];
            dist[i]=dist[j];
            dist[j]=temp;
        }
    }
    cout<<"\nThe smallest distance is\n";
    cout<<dist[0]<<endl;
    cout<<"\nThe largest distance is\n";
    cout<<dist[m-1]<<endl;
return 0;
}

```

output:

Please make sure that all the points you enter are distinct

Enter the number of points

4

Enter the points row-wise

1 2

3 4

5 6

7 8

(1,2)

(3,4)

The distances are: dist[0]2.82843

(1,2)

(5,6)

The distances are: dist[1]5.65685

(1,2)

(7,8)

The distances are: dist[2]8.48528

(3,4)

(5,6)

The distances are: dist[3]2.82843

(3,4)

(7,8)

The distances are: dist[4]5.65685

(5,6)

(7,8)

The distances are: dist[5]2.82843

The smallest distance is
2.82843

The largest distance is
8.48528

Process exited after 35.23 seconds with return value 0
Press any key to continue . . .